

Software Tester (QA) Assessment

Submitted by: Akshit Bharti

Full Name	Akshit Bharti
Email ID	akshit.bharti.aiml.2023@mitmeerut.ac.in
Contact Number	7310932476

1. Test Case Design — Registration

TC ID	Test Case	Expected Result	Type
REG-01	Register with valid name, unique email and valid password.	Account should be created successfully.	Positive
REG-02	Register with an already registered email.	An appropriate error should be displayed.	Negative
REG-03	Leave name blank and submit.	Required-field validation should appear.	Negative
REG-04	Leave email blank and submit.	Required-field validation should appear.	Negative
REG-05	Leave password blank and submit.	Required-field validation should appear.	Negative
REG-06	Enter an invalid email format.	Invalid email message should appear.	Negative
REG-07	Enter a weak/invalid password.	Password validation should appear.	Negative
REG-08	Test the minimum allowed password length.	Minimum valid length should be accepted.	Edge
REG-09	Enter one character less than the minimum length.	Registration should be rejected.	Edge
REG-10	Enter very long input values.	Application should handle them without crashing.	Edge
REG-11	Enter special characters in input fields.	Input should be handled correctly.	Edge
REG-12	Enter only spaces in a required field.	Input should not be accepted.	Negative

2. Test Case Design — Login

TC ID	Test Case	Expected Result	Type
LOG-01	Login with valid registered credentials.	User should successfully log in.	Positive
LOG-02	Enter correct email with wrong password.	Login should fail with a proper error.	Negative
LOG-03	Enter an unregistered email.	Login should fail.	Negative
LOG-04	Leave email blank.	Email validation should appear.	Negative
LOG-05	Leave password blank.	Password validation should appear.	Negative
LOG-06	Enter an invalid email format.	Email validation should appear.	Negative
LOG-07	Test different letter cases in credentials.	Behavior should follow the application's requirements.	Edge
LOG-08	Enter wrong credentials repeatedly.	Repeated attempts should be handled securely.	Edge
LOG-09	Enter malicious input in login fields.	Input should be safely handled; no unauthorized access.	Security
LOG-10	Login, logout, then use browser Back.	Protected pages should not become accessible.	Edge

3. Test Case Design — Task CRUD

Create Task

TC ID	Test Case	Expected Result	Type
TASK-C01	Create a task with valid details.	Task should be created and displayed.	Positive
TASK-C02	Create a task with an empty title.	Validation message should appear.	Negative
TASK-C03	Create a task with only spaces.	Task should not be created.	Negative
TASK-C04	Create the same task twice.	Application should follow the defined duplicate-task behavior.	Edge
TASK-C05	Enter a very long task title.	Application should handle the input correctly.	Edge
TASK-C06	Enter special characters.	Task should be stored and displayed correctly.	Edge
TASK-C07	Try saving when the database is unavailable.	A proper error should appear; no false success.	Negative

View Task List

TC ID	Test Case	Expected Result	Type
TASK-V01	Open the task list with existing tasks.	User's tasks should be displayed.	Positive
TASK-V02	Open the task list when no tasks exist.	A clear empty-state message should appear.	Edge
TASK-V03	Create a task and open the list.	The new task should appear.	Positive
TASK-V04	Refresh the task list.	Saved tasks should remain available.	Positive
TASK-V05	Login as another user and inspect the list.	Other user's tasks should not be visible.	Security
TASK-V06	Use a large number of tasks.	The list should load without crashing.	Edge

Edit Task

TC ID	Test Case	Expected Result	Type
TASK-E01	Edit an existing task and save.	Updated task should be displayed.	Positive
TASK-E02	Remove the title and save.	Validation should prevent an invalid update.	Negative
TASK-E03	Enter a very large value while editing.	Application should handle it correctly.	Edge
TASK-E04	Edit a task and then cancel.	Original task should remain unchanged.	Positive
TASK-E05	Attempt to edit a task that does not exist.	Proper error should be displayed.	Negative
TASK-E06	Attempt to edit another user's task.	Access should be denied.	Security

Delete Task

TC ID	Test Case	Expected Result	Type
TASK-D01	Delete an existing task.	Task should be removed.	Positive
TASK-D02	Select Delete and then Cancel.	Task should remain.	Positive
TASK-D03	Select Delete and Confirm.	Task should be removed from the list/database.	Positive
TASK-D04	Attempt to delete a task that does not exist.	Proper error should be handled.	Negative
TASK-D05	Attempt to delete another user's task.	Operation should be rejected.	Security
TASK-D06	Delete a task and refresh the page.	Deleted task should not reappear.	Edge

4. Input Validation & Error Handling

TC ID	Scenario	Expected Result
VAL-01	Required field left blank.	Clear validation message.
VAL-02	Invalid email.	Email format error.
VAL-03	Input contains only spaces.	Input should be rejected.
VAL-04	Input exceeds maximum length.	Validation/error without application crash.
VAL-05	Special characters entered.	Input should be properly validated/escaped.
VAL-06	Database unavailable.	User-friendly error message.
VAL-07	Server/API error.	Meaningful error; no technical stack trace.
VAL-08	Network interruption.	Failure should be handled gracefully.
VAL-09	Unauthorized access.	Access should be denied or user redirected to login.
VAL-10	Invalid task ID.	Appropriate error instead of application crash.
VAL-11	Repeated button click.	Should not create duplicate tasks/requests.
VAL-12	Session expires.	User should be asked to log in again.

5. Potential Bugs / Risk Areas

#	Potential Bug / Risk	Severity	Reason / Impact
1	User can access another user's tasks.	Critical	Serious privacy and data-security issue.
2	Password stored insecurely in the database.	Critical	A database compromise could expose user passwords.
3	SQL injection through login/task fields.	Critical	Could allow unauthorized database access or manipulation.
4	Task deletion happens without confirmation.	Major	Users could accidentally delete important tasks.
5	Created/edited task is not saved correctly.	Major	Users may lose task information after refresh.
6	Deleted task reappears after refresh.	Major	Shows a database persistence/synchronization problem.
7	Invalid/empty task can be created.	Major	Creates unusable or incomplete task records.
8	Session remains active after logout.	Major	Another person could access the account on the device.
9	Application crashes on very long input.	Major	Can affect application stability.
10	Poor handling of database/server failures.	Minor	Users may receive confusing errors or false success.

6. Recommended Test Execution Priority

1. Authentication and login
2. Authorization and user-data isolation
3. Create task
4. Edit task
5. Delete task
6. Database persistence
7. Input validation
8. Error handling
9. Logout/session security
10. Edge cases and UI behavior

QA Conclusion

Before production release, I would not approve the application based only on happy-path testing. Since there is no existing test documentation, I would first create a core regression suite and focus on authentication, authorization, database persistence, CRUD operations, validation, error handling and security.